Distance Meter with LCD

Arduino Uno R3 + Tinkercad | HC-SR04 Ultrasonic Sensor + 16x2 LCD Display

Yeh project HC-SR04 ultrasonic sensor use karke distance measure karta hai aur result seedha **16x2 LCD screen** par live show karta hai. Code is PDF mein copy-paste ke liye maujood hai, aur sath mein alag **Distance_Meter_LCD.ino** file bhi di gayi hai jo seedha Tinkercad ya Arduino IDE mein open ki ja sakti hai.

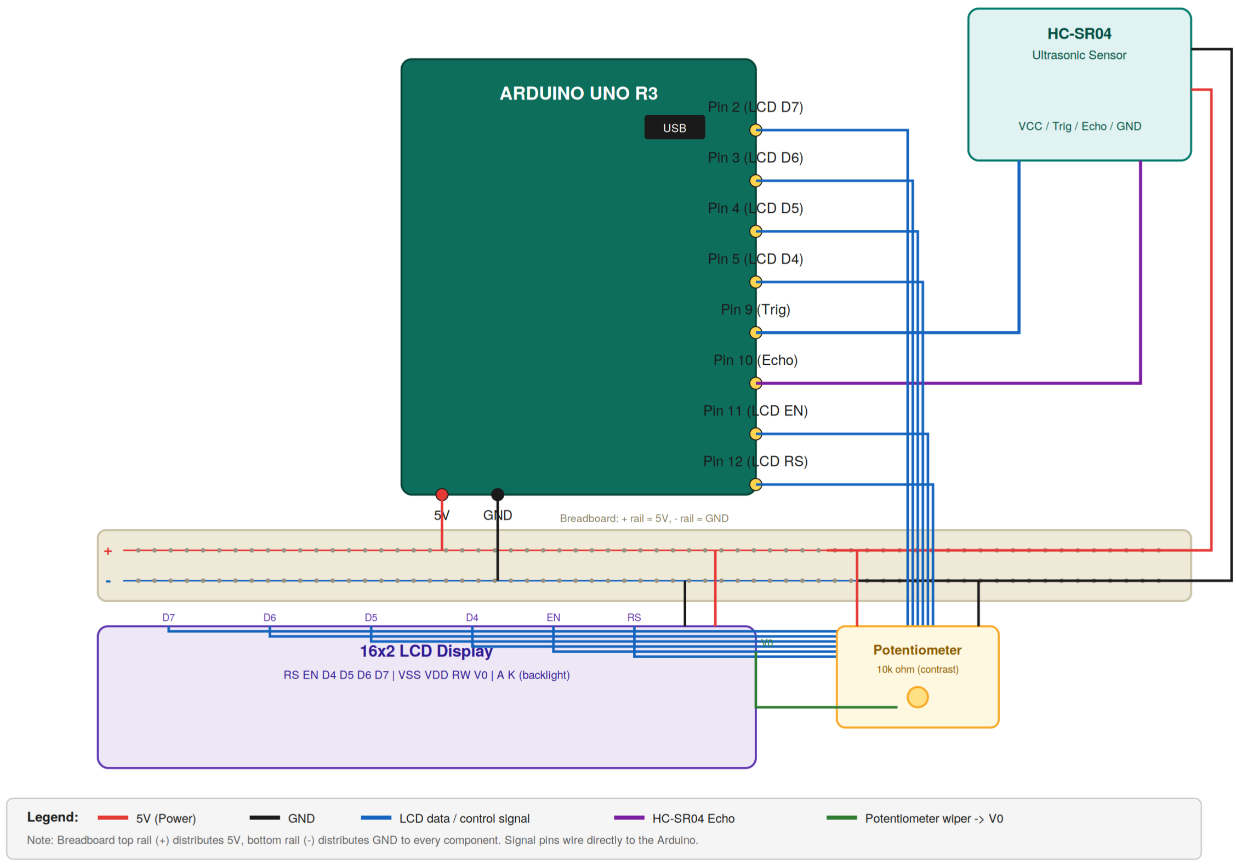
1. Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller board
HC-SR04 Ultrasonic Sensor	1	Measures distance using sound waves
16x2 LCD Display	1	Shows live distance reading
Potentiometer (10k ohm)	1	Adjusts LCD contrast
Breadboard	1	For connections
Jumper Wires	As needed	For wiring all components

2. Circuit Diagram

White background par color-coded wiring niche di gayi hai (legend diagram ke neeche hai):

Distance Meter with LCD — Wiring Diagram (Breadboard Layout)



3. Build Steps

1. Tinkercad open karo aur naya circuit banao.
2. Arduino Uno, breadboard, HC-SR04, 16x2 LCD, aur potentiometer sab drag karke place karo.
3. HC-SR04 wiring: VCC to 5V, GND to GND, Trig to Pin 9, Echo to Pin 10.
4. LCD wiring: RS to Pin 12, EN to Pin 11, D4 to Pin 5, D5 to Pin 4, D6 to Pin 3, D7 to Pin 2, VSS to GND, VDD to 5V, RW to GND, backlight A to 5V, backlight K to GND.
5. Potentiometer wiring: dono ends 5V aur GND se, middle pin (wiper) LCD ke V0 se.
6. Code editor open karo aur LiquidCrystal.h library wala code paste karo.
7. Simulation run karo — LCD screen par live distance dikhna chahiye.

4. Full Arduino Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

#define trigPin 9
#define echoPin 10

void setup() {
  lcd.begin(16, 2);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  lcd.print("Distance Meter");
  delay(1500);
  lcd.clear();
}

void loop() {
  long duration, distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  lcd.setCursor(0, 0);
  lcd.print("Distance:");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm   ");

  delay(500);
}
```

Tip: Is code ko seedha copy-paste kar sakte ho, ya sath diya gaya **Distance_Meter_LCD.ino** file directly Tinkercad/Arduino IDE mein open kar lo.

5. Wiring Summary Table

Component Pin	Full Name	Arduino Pin	Pin Type	Wire Color
HC-SR04 - VCC	Voltage Common Collector	5V	Power	Red
HC-SR04 - GND	Ground	GND	Ground	Black
HC-SR04 - Trig	Trigger	Pin 9	Digital Output	Blue
HC-SR04 - Echo	Echo	Pin 10	Digital Input	Purple
LCD - RS	Register Select	Pin 12	Digital Output	Blue
LCD - EN	Enable	Pin 11	Digital Output	Blue
LCD - D4	Data Bus Line 4	Pin 5	Digital Output	Blue

Component Pin	Full Name	Arduino Pin	Pin Type	Wire Color
LCD - D5	Data Bus Line 5	Pin 4	Digital Output	Blue
LCD - D6	Data Bus Line 6	Pin 3	Digital Output	Blue
LCD - D7	Data Bus Line 7	Pin 2	Digital Output	Blue
LCD - VSS / RW	Ground / Read-Write	GND	Ground	Black
LCD - VDD	Voltage Drain Supply	5V	Power	Red
LCD - V0	Contrast Voltage	Potentiometer wiper	Contrast Control	Green
LCD - Backlight A	Anode	5V	Power	Red
LCD - Backlight K	Cathode	GND	Ground	Black
Potentiometer - Ends	Power Terminals	5V / GND	Power	Red / Black

Note: Arduino Uno ke Pin 0 aur Pin 1 serial communication ke liye reserved hain, isliye is project mein use nahi kiye gaye. Wire colors upar circuit diagram ke legend se match karte hain — isse Tinkercad mein circuit banate waqt sahi color ki jumper wire choose karna asaan ho jata hai.

6. Kaise Kaam Karta Hai

- HC-SR04 wahi tarike se distance measure karta hai — Trig pulse bhejna, Echo pulse ka duration parhna, aur formula se cm mein convert karna.
- Result Serial Monitor ki bajaye **LiquidCrystal.h** library ke zariye 16x2 LCD screen par likha jata hai.
- **lcd.begin(16, 2)** LCD ko initialize karta hai (16 columns, 2 rows). **lcd.setCursor(col, row)** batata hai ke agla text kahan likhna hai, aur **lcd.print()** asal value LCD par display karta hai.
- Potentiometer LCD ke V0 pin se juda hota hai taake screen ka contrast manually adjust kiya ja sake — agar text nazar na aaye to pot ko ghuma kar theek kiya jata hai.
- RW pin ko direct GND se joda gaya hai kyunki hum LCD ko hamesha 'write mode' mein rakhte hain (kabhi LCD se data parhna nahi hota).
- Har 500 milliseconds mein loop() naya distance measure karta hai aur LCD ki dusri row update karta rehta hai, isliye reading real-time mehsoos hoti hai.